

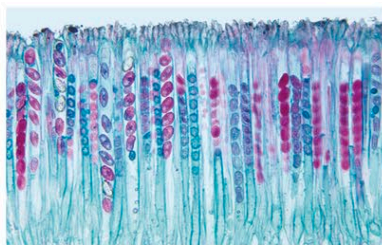
SAC FUNGI

The Ascomycota, or sac fungi, are fungi that produce their spores in rounded to elongate sacs called asci located on the fertile surface of the fruitbody. They are the largest group of fungi, and include many cup- or saucer-shaped species.

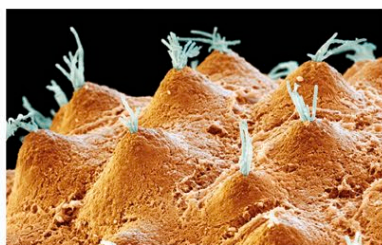
PHYLUM	ASCOMYCOTA
CLASSES	7
ORDERS	56
FAMILIES	226
SPECIES	About 33,000



Many sac fungi display vivid colours, although it is uncertain what biological function such bright colours may have.



Spore-producing asci are shown here under a microscope. Arranged in dense layers, the asci contain eight spores each.



Many species form their asci in special protective chambers called perithecia; from here they discharge their spores.

DEBATE

HEROES OR VILLAINS?

Sac fungi form beneficial associations with plants, algae, and even arthropods such as beetles. But equally they include some of the world's most damaging pathogens: *Cryphonectria*, for example, is responsible for the recent death of millions of chestnut trees. Perhaps no other group of fungus has such contrasting effects on our world.

Ranging in size from the microscopic to around 20 cm (8 in) in height, the sac fungi occupy a wide variety of habitats, growing on dead, dying, and living tissue, and floating in fresh and saltwater environments. Many species are parasitic, and include some of the most serious crop pests. Others form mutually beneficial relationships, called mycorrhizae, with plants. Sac fungi include some of the most important fungi in the history of medicine, such as the source of penicillin, while others are serious pathogens – *Pneumocystis jirovecii*, for instance, can cause lung infections in people with a weak immune system. This phylum also contains the yeasts, which, being vital components in the production of alcohol and bread, have played a pivotal role in human history.

The Ascomycota display a wide range of fruitbody shapes, including cup-shaped, club-shaped, potato-like, simple crusts or sheets, pimple-like, coral-like, shield-shaped, or stalked with a sponge-like cap. Depending on the type of fruitbody, the spore-producing asci may grow externally on a special fertile layer or be contained internally. Not all species have a sexual stage; in fact, many use asexual methods of reproduction. Most yeasts grow and rapidly colonize new areas as a result of asexual division, or budding, where a small bud forms on the outside of a yeast cell, then separates, becoming a new cell.

THE CUP FUNGUS

The common name of cup fungus refers to one of the most conspicuous shapes of fruitbody associated with the Ascomycota. Its open top (which is sometimes disc- or saucer-shaped) allows wind and rain to scatter the spores that line its inner surface. In some varieties, water is absorbed by the asci, which contain the spores, so that pressure builds up and forcibly ejects the spores up to 30 cm (12 in) from the fruitbody. Close examination of surfaces such as rotting logs, fallen branches, or leaves may reveal a fascinating world of almost microscopic cups. In the larger cup species, disturbance of the cup can produce such a vigorous ejection of spores that their release is not only visible as a faint cloud of spores, but also audible.

ORANGE PEEL CUP >

The orange peel fungus is a good example of the simple cup-shaped fruitbody adopted by many fungi species.

